

Abheer Shambharkar

Game Programmer and Designer

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PROFILE

Game programmer and technical designer with a software engineering background, currently completing an MSc in Computer Games at Queen Mary University of London. Solo-designed and built Yassantra, a horror game centered on a custom-built enemy AI covering the full finite-state-machine architecture and sensor systems, as well as level layout, encounter pacing and environmental storytelling. Comfortable owning both sides of a feature: the systems that make an AI or mechanic work along with the design decisions that make it worth building. Also led environment design, enemy balancing and narrative direction on FeatherBound, a university team project and completed a professional internship at Elyndor Interactive owning AI and level implementation on an unreleased title. Background spans web development and software engineering prior to moving into games full-time.

PROFESSIONAL EXPERIENCE

Programming Intern, Elyndor Interactive

December 2025 – June 2026 | Remote

Joined as a programming intern on *Kaali: A Mother's Tale*, an unreleased horror title, at an early-stage startup studio.

- Took full ownership of one of the game's starting levels, handling implementation end-to-end
- Partway through the internship, identified that the project's workflow was poorly structured and restructured it, improving how work was organized and tracked for the rest of the team
- Assumed informal assistant-lead responsibilities, contributing to design decisions, scope definition and timeline planning as the studio lacked an established structure for this
- In the final phase, took ownership of the enemy AI system as a whole, working from a rough behavioral brief. Designed and implemented the full detection system, including vision sensors and configurable detection range, expanding a simple brief into a properly refined AI system
- Delivered complete, working AI scripts to the team before the internship concluded

Web Designing and Development Intern

September 2024 – February 2025 | Nagpur, India

AbCoEdTech Pvt. Ltd. 

- Built the Rahbar Badminton Academy website, a full front-end and back-end build using HTML5, CSS, Bootstrap, SQL, PHP and JavaScript with a PHP/SQL backend for dynamic content

Web Developer Intern, AbCoEdTech Pvt. Ltd.

January 2024 – May 2024 | Nagpur, India

- Built the National Institute of Fire Safety Engineering landing page, a static site using plain HTML5 and CSS

PROJECTS

Yassantra , 3D horror game

July 2025

Kaali: A Mother's Tale is an unreleased title where Yassantra's foundational AI system originated during my Elyndor internship, Yassantra independently redesigns and expands that system into a complete game in around 20 days.

- Designed and implemented a 6-state enemy AI finite state machine (Patrol, Alert, Chase, Attack, Search, custom-built Investigate state) running on Unity NavMesh
- Rebuilt the inherited hearing sensor from a hard-capped model into a fully sphere-based, uncapped detection system supporting long-range triggers
- Built a unified, event-driven sound propagation system covering every audio source in the game (player movement, screams, alarm clocks, tape recorder, hazard explosions, siren) with per-source tunable radii
- Designed and implemented a global trap counter enforcing a session-wide cap on active traps, with capacity freed as traps are destroyed or disarmed
- Built dual-zone trap interactions (disarm vs. trigger), hold-based disarm/escape mechanics, auto-release timers and a separate fuse-based hazard system with tiered proximity damage
- Designed and implemented a 3-gate escape sequence (Main Gate, Shut Gate, End Gate) with siren-triggered AI rerouting, corridor detection logic and a genuine fail-state path if the player mismanages the sequence
- Built a single SphereCast-based interaction framework from the player camera, driving all world interactions across items, hangers, traps and beacons

- Implemented the full item system (torch, alarm clocks, health potions, box key, tape recorder) with key-press activation and win condition checks
- Built the HUD, pause menu, settings, death screen and full scene reset logic on restart
- Solo developer across every system above — programming, design, level layout, atmosphere, lighting and UI using Unity Asset Store assets only for select visual/audio elements

FeatherBound *🔗*, 3D Survival Adventure Game

October 2025 – January 2026

- Owned lore, game direction and overall narrative framing for the project
- Designed and built all four biome environments (Skyaera, Hailcrest, Scorchvaal, Darkturnum), including terrain layout, environmental storytelling and placement of resources, enemies and points of interest
- Designed and implemented enemy behaviour and balancing values across all biomes, including difficulty scaling for later regions
- Authored the initial game concept and environment design on paper before implementation, iterating from an early prototype that was scaled back after being flagged as too complex for the milestone
- Collaborated with two team members who handled player systems, UI and core gameplay loop implementation

Galaxy Shooter *🔗*, 2D Arcade Shooter Game

November 2025 – January 2026

- Built the core game using GameDevHQ's Unity course as the foundational framework and asset base
- Designed and implemented an original difficulty-scaling system: enemy movement speed accelerates continuously over the first 200 seconds, reaching 3x starting speed, rather than remaining constant
- Implemented three collectible power-ups (Shield, Triple Shot, Speed Boost) and a three-life system inspired by classic arcade design
- Built local high score tracking and quick-restart functionality

AirVOID V-1.2 *🔗*, 2D Endless Arcade Game

September 2024

- Followed Game Maker's Toolkit's Flappy Bird clone tutorial to build the core game loop, using the associated free asset pack
- Implemented menu and in-game exit/quit functionality on top of the base tutorial
- Added out-of-bounds death detection so the bird cannot exploit screen boundaries to bypass pipes
- Handled full project setup and configuration in Unity 2021.3 LTS, marking first hands-on experience with the engine

Demo Portfolio *🔗*, Experimental Portfolio Website

November 2024 – December 2024

- An earlier dummy version of my portfolio site, built with plain HTML5 and CSS before the current content.
- Designed and built the entire site solo using HTML5 and CSS

Optimized College Placement System *🔗*, Web Application

September 2023 – July 2024

- A placement management system built as my major college project, using a full web stack to handle student and recruiter data, listings and placement workflows.
- Built both front end and back end using HTML5, CSS, Bootstrap, SQL, PHP and JavaScript with team.

Rahber Badminton Academy *🔗*, Dynamic Website and Web App

September 2024 – February 2025

- Full website built during my web design and development internship, featuring a PHP and SQL backend for dynamic content and data handling.
- Built both front end and back end using HTML5, CSS, Bootstrap, SQL, PHP and JavaScript

Nagpur Institute of Fire Safety Engineering *🔗*, Landing Web Page

January 2024 – May 2024

- A static landing page built during my first web development internship, using plain HTML5 and CSS with no backend framework.
- Designed and built the site solo using HTML5 and CSS

TECHNICAL SKILLS

Programming

C, C++, C#, Java, Python, PHP, Javascript and SQL

AI & Systems

Finite state machines, NavMesh pathfinding, sensor-based detection (vision/hearing), event-driven sound propagation, global resource/session counters, interaction frameworks

Web Development

HTML5, CSS, Bootstrap 5, XAMPP (local backend development)

Office Suite

Word, PowerPoint, Excel, To Do and Outlook

Tools

Git/GitHub, Visual Studio, Figma, Canva, PicsArt

LONG-TERM PERSONAL CHALLENGE

Game Alphabet Development (GAD) Series

June 2026 – Present

A challenge for myself to build 26 games, one per letter of the alphabet, developed out of order based on whichever skill, genre or technology I want to explore next. Four entries completed to date:

- AirVOID (Project A — arcade fundamentals)
- Galaxy Shooter (Project G — gameplay systems and balancing)
- FeatherBound (Project F — large-scale survival, team collaboration)
- Yassantra (Project Y — AI-driven horror, solo technical showcase).

EDUCATION

MSc in Computer Games

Queen Mary University of London 

September 2025 – September 2026
London, United Kingdom

B.Tech in Computer Science and Engineering

Rashtrasant Tukadoji Maharaj Nagpur University 

2020 – 2024 | Nagpur, India

Higher Secondary Education and IT, St. Paul Junior College

2018 – 2020 | Nagpur, India

Secondary Education, Sri Guru Harkrishan Public School

2009 – 2018 | Nagpur, India

EXTRA COURSES

The Ultimate Guide to Game Development with Unity , Udemy

Unity Certified Official Course

2024 – 2026 | Virtual

Maharashtra State Certificate in Information Technology , Inline Computer

Foundational computer literacy course on MS Office and essential digital skills

2015 | Nagpur, India

ORGANISATIONS

Queen Mary University of London

Course Representative

October 2025 – Present | London, United Kingdom

- Represent student feedback to academic staff
- Managing this alongside a full-time MSc programme

Jhulelal Institute of Technology, Training & Placement Department Coordinator

2021 – 2024 | Nagpur, India

- Coordinated placement drives and recruitment events
- Arranged meetings between students and faculty
- Coordinated company representatives.